SYSTEM ARCHITECTURE • EXECUTIVE BLUEPRINT

Architectural Paradigm: Traditional ERPs vs Our Platform

An empirical contrast between legacy monolithic school software and our Next.js 15 multi-tenant, offline-first operating system.

⚠ How Traditional ERPs Are Built

- ✕ **Legacy Tech Stack**: Built on ASP.NET, PHP 5/7, or Java/Oracle from the 2000s; heavy desktop client remnants and slow jQuery spaghetti.
- ✕ **Single-Tenant Monolith**: Each school runs a separate database instance or manual server; adding a new institute takes weeks of manual IT setup.
- ✕ **Zero Offline Capability**: Hard-coupled to live server connections; if school Wi-Fi or broadband drops, classroom operations instantly freeze.
- ✕ **Rigid Hierarchy**: Hardcoded for standard schools only; cannot model universities, semester credit banks, or coaching batches.
- ✕ **Cluttered 90s UI**: Overwhelming forms with 50+ dropdowns, zero mobile optimization, and painful multi-click navigation.

✦ What Sets Our Platform Apart (Unfair Advantages)

- ✓ **Next.js 15 App Router & React 19**: Blindingly fast server-rendered pages with sub-100ms client transitions and strict TypeScript type-safety.
- ✓ **Shared Database with PostgreSQL RLS**: Instant 200ms tenant provisioning with kernel-level cryptographic tenant isolation via Row-Level Security.
- ✓ **True Offline-First Engine (Dexie.js)**: Classroom attendance & mark entry cache in browser IndexedDB; mutations queue in durable Outbox and auto-sync on reconnect.
- ✓ **Adaptive 4-in-1 Terminology**: Single codebase morphs into a School, College, University, or Coaching Academy ERP without code duplication.
- ✓ **Linear/Notion Consumer Polish**: Universal Command Palette (`Ctrl + K`), Driver.js Guided Tours, and high-contrast WCAG AA Dark/Light modes.

Core Architectural & Platform Governance Features (Module 1 & 2 Sample)

All controlled via tenant feature flags

ID	FEATURE NAME	TECHNICAL FUNCTION & CAPABILITY	PRIORITY	FEATURE FLAG KEY
F001	**Subdomain Auto-Provisioning**	Instant subdomain routing (`{slug}.edusaas.com`) executed via Next.js Middleware.	P0 Core	`feat_subdomain`
F002	**Custom Domain CNAME Binding**	Host header mapping supporting verified custom domains with live DNS resolution.	P0 Core	`feat_custom_domain`
F003	**Automated SSL Issuance**	Zero-touch Let's Encrypt / Cloudflare for Platforms automated SSL certificate generation.	P0 Core	`feat_auto_ssl`
F013	**PostgreSQL Row-Level Security**	Database kernel-level security policies preventing cross-tenant data leakage.	P0 Core	`feat_tenant_rls`
F026	**Global Tenant Registry**	Super Admin searchable directory of all registered institutes, health, and student counts.	P0 Core	`feat_saas_core`
F029	**Per-Tenant Feature Flag Matrix**	Interactive UI matrix allowing Super Admin to toggle any of the 580 features per tenant.	P0 Core	`feat_flag_matrix`
F030	**Audited Instant Impersonation**	1-Click login into any tenant account with cryptographic SHA-256 audit log tracking.	P0 Core	`feat_audit_impersonation`

MODULE 1 & 3 • WHITE-LABELING & ONBOARDING

Multi-Tenancy, Self-Service Onboarding & Custom Domains

Enabling school, college, and academy owners to launch their own branded portal in under 10 minutes without writing a single line of code.

⚠ Generic ERP Domain & Onboarding Model

- ✕ **4–12 Weeks Implementation**: School must wait for vendor engineering teams to manually deploy code and configure database instances.
- ✕ **Vendor-Locked Domains**: Schools are forced to use ugly vendor links (e.g. `vendor.com/school/login.php?id=492`).
- ✕ **Exorbitant White-Label Charges**: Vendors charge ₹1 Lakh to ₹2 Lakhs upfront just to put the school's logo on the login page.
- ✕ **Fixed Colors & Fonts**: Generic blue or grey UI that cannot be matched to the school's official brand colors.

✦ Our Instant White-Label & Onboarding Engine

- ✓ **10-Minute Guided Wizard**: Self-service flow: Name → Archetype → Branding → Domain → Modules → 1-Click Launch.
- ✓ **Instant Custom Domain CNAME**: Map `portal.dpsdelhi.in` with automated DNS verification and instant Let's Encrypt SSL.
- ✓ **Dynamic CSS Token Injection**: Primary/secondary colors, fonts, and dark/light logos injected directly into CSS variables at runtime.
- ✓ **1-Click Demo Data Seeder**: Allows school owners to evaluate the portal with pre-populated sample students and classes in 5 seconds.

Onboarding, Domains & White-Labeling Feature Catalogue (Sample)

Module 1 & 3 Full Specifications

ID	FEATURE NAME	TECHNICAL FUNCTION & CAPABILITY	PRIORITY	FEATURE FLAG KEY
F004	**Dynamic Tenant Theme Injection**	Server-rendered CSS custom properties (`--tenant-primary`) injected per tenant host.	P0 Core	`feat_custom_branding`
F007	**White-Labeled Login Portal**	Dedicated login screen featuring institute logo, motto, badges, and background styling.	P0 Core	`feat_whitelabel_login`
F009	**Custom SMTP Sender Identity**	Transactional emails dispatched directly through institute's own GSuite/Office365 SMTP.	P0 Core	`feat_custom_smtp`
F010	**White-Label SMS Sender Header**	Dispatches SMS using the institute's TRAI DLT approved 6-character header (e.g. `DPSDEL`).	P0 Core	`feat_sms_sender_id`
F046	**10-Minute Guided Onboarding Wizard**	6-step interactive onboarding with live preview cards and validation diagnostics.	P0 Core	`feat_onboarding_wizard`
F049	**Enabled User Types Selector**	Admin toggles whether Student and Parent logins are activated or hidden entirely.	P0 Core	`feat_user_type_toggle`
F052	**1-Click Demo Data Seeder**	Pre-populates sample classes, teachers, students, and marks for instant trial evaluation.	P0 Core	`feat_demo_seeder`

MODULE 4 • CURRICULUM & STRUCTURE

Adaptive Terminology Transformer & NEP 2020 Curriculum

Dynamic organizational hierarchy adapting seamlessly between K-12 Schools, Degree Colleges, Universities, and Coaching Academies.

⚠ Generic ERP Academic Limitations

- ✕ **Rigid K-12 Only Model**: Built only for "Class 10-A". When a college tries to use it, it cannot represent Degrees, Semesters, or Specializations.
- ✕ **No Credit System**: Incapable of weighting courses by credit hours (e.g. 4 credits Theory, 2 credits Practical).
- ✕ **No NEP 2020 Support**: Zero support for Multiple Entry/Exit points, Academic Bank of Credits (ABC), or Major/Minor multidisciplinary streams.
- ✕ **Hardcoded Labels**: Cannot change "Section" to "Batch" or "Class Teacher" to "Mentor" without modifying database source code.

✦ Our Adaptive Terminology & NEP Engine

- ✓ **Semantic Terminology Hook**: React `useTerminology()` dynamically renders exact labels based on active tenant archetype.
- ✓ **NEP 2020 Multiple Entry/Exit**: Tracks student credits for Certificate (Yr 1), Diploma (Yr 2), Degree (Yr 3), and Honours (Yr 4).
- ✓ **Academic Bank of Credits (ABC)**: Native storage and export formatting for 12-digit DigiLocker APAAR / ABC IDs.
- ✓ **Major, Minor & Electives**: Flexible course allocation allowing students to pick open electives and skill enhancement courses.

Academic Hierarchy & Curriculum Feature Catalogue (Sample)

Module 4 Full Specifications

ID	FEATURE NAME	TECHNICAL FUNCTION & CAPABILITY	PRIORITY	FEATURE FLAG KEY
F048	**Dynamic Terminology Transformer**	Dynamically translates UI terms (Class vs Program, Section vs Semester, Teacher vs Mentor).	P0 Core	`feat_terminology_engine`
F066	**Multi-Year Academic Session Manager**	Manage active, past, and upcoming academic sessions with 1-click active session switching.	P0 Core	`feat_academic_sessions`
F070	**NEP 2020 Major & Minor Tracks**	Tags subjects as Major, Minor, Multidisciplinary, or Skill Enhancement per NEP guidelines.	P0 Core	`feat_nep_curriculum`
F072	**Credit Hour Weighting System**	Assigns academic credits per subject for UGC/AICTE degree audit and GPA calculations.	P0 Core	`feat_credit_system`
F073	**Elective Subject Grouping**	Allows students to pick elective courses with automated clash-validation rules.	P0 Core	`feat_elective_selection`
F086	**Practical Lab Group Batches**	Splits a section into Batches (B1, B2, B3) for laboratory sessions and clinical rotations.	P0 Core	`feat_lab_batches`
F088	**Academic Session Rollover Wizard**	Automated 1-click duplication of academic structure into the upcoming academic year.	P0 Core	`feat_session_rollover`

MODULE 5 • ADMISSIONS & DIRECTORY

Student Admissions, Identity Rules & Lifecycle Management

Complete student journey from online inquiry to 60-field profile, automated identity numbering, bulk CSV import, and Transfer Certificates.

⚠ Generic ERP Student Management

- ✕ **Limited Basic Fields**: Collects only name, father's name, and phone; misses statutory Indian caste categories and blood groups.
- ✕ **No Indian Identity Numbers**: Cannot handle University PRN (Permanent Registration No), APAAR ID, or UDISE Student PEN.
- ✕ **Painful Bulk Import**: Fails silently on CSV errors with zero preview or row-by-row error highlighting.
- ✕ **Manual Transfer Certificates**: Staff type TCs in MS Word manually, leading to duplicate numbers and zero security verification.

✦ Our Modern Student Lifecycle Platform

- ✓ **60+ Field Profile Vault**: Encrypted storage of personal, medical, emergency, and guardian demographic data.
- ✓ **Automated Identity Numbering**: Customizable auto-increment rules for Admission No, Section Roll No, and University PRN.
- ✓ **Interactive CSV Validation Grid**: PapaParse streaming parser highlights exact row errors with 1-click import rollback.
- ✓ **Government Formatted TC Generator**: Checks fee and library clearance; auto-prints security QR verified Transfer Certificate.

Student Admissions & Lifecycle Feature Catalogue (Sample)

Module 5 Full Specifications

ID	FEATURE NAME	TECHNICAL FUNCTION & CAPABILITY	PRIORITY	FEATURE FLAG KEY
F094	**Unique Admission No Generator**	Auto-generates customizable scholar numbers (e.g. `DPS/2026/0491`) with collision locks.	P0 Core	`feat_admission_numbering`
F096	**Indian PRN & Enrollment Tracking**	16-digit university Permanent Registration Number tracking for lifelong transcripts.	P0 Core	`feat_prn_tracking`
F097	**DigiLocker APAAR / ABC ID Field**	Stores and validates the 12-digit National Automated Permanent Academic Account Registry ID.	P0 Core	`feat_apaar_id`
F098	**Face-Detection Photo Cropper**	In-browser canvas tool auto-cropping student portraits to passport photo dimensions.	P0 Core	`feat_photo_cropper`
F100	**Sibling Linking Engine**	Links brothers and sisters across classes to automatically trigger fee concession rules.	P0 Core	`feat_sibling_linking`
F104	**RTE 25% Statutory Quota Flag**	Flags Right to Education students with automatic 100% tuition fee billing waivers.	P0 Core	`feat_rte_quota`
F109	**Transfer Certificate (TC) Generator**	Automated official TC generation with school seal, conduct remarks, and security QR.	P0 Core	`feat_tc_generator`

MODULE 6 • REGULATORY COMPLIANCE • LEGAL IMMUNITY

Indian Statutory Compliance: RTE, UDISE+, MoE 2024 & DPDPA

Built-in legal protection adhering strictly to the Digital Personal Data Protection Act 2023 and Ministry of Education coaching regulations.

⚠ Legal Vulnerabilities in Generic ERPs

- ✕ **Violates DPDPA 2023**: Embeds commercial third-party trackers (Google Ads, Facebook Pixels) tracking minor children's online behavior.
- ✕ **Coaching Centers at Legal Risk**: Unaware of the 2024 MoE rules; allows enrolling students under 16 and lacks the mandatory pro-rata refund calculator.
- ✕ **No CERT-In Log Retention**: Deletes web access logs after 7–30 days, violating the mandatory 180-day retention directive.
- ✕ **Manual UDISE+ & AISHE**: Clerks spend 3 weeks manually compiling government data spreadsheets, prone to reporting fines.

✦ 100% Legal Immunity in Our Architecture

- ✓ **DPDPA Section 9 Minor Shield**: Verifiable Parental Consent via OTP; zero commercial tracking; zero advertising directed at children.
- ✓ **MoE 2024 Coaching Legal Engine**: Under-16 admission validation blocker + automated 10-day statutory pro-rata refund calculator.
- ✓ **180-Day Tamper-Proof Audit Logs**: Immutable database logs retained in Indian cloud data centers complying with CERT-In directives.
- ✓ **1-Click Statutory Government Exports**: Direct CSV export matching official SDMIS (UDISE+) and DCF-II (AISHE) specifications.

Indian Statutory Compliance Feature Catalogue (Sample)

Module 6 Full Specifications

ID	FEATURE NAME	TECHNICAL FUNCTION & CAPABILITY	PRIORITY	FEATURE FLAG KEY
F126	**UDISE+ School Profile Master**	Maintains School UDISE Code, Latitude, Longitude, Rural/Urban and School Category tags.	P0 Core	`feat_udise_master`
F127	**UDISE+ SDMIS Export Engine**	1-Click export of student records formatted exactly per Ministry SDMIS specifications.	P0 Core	`feat_udise_sdmis_export`
F128	**AISHE College DCF-II Export**	Higher education institutional data compilation for Ministry of Education reporting.	P0 Core	`feat_aishe_export`
F129	**Statutory Social Category Roster**	Real-time count of SC, ST, OBC (Creamy/Non-Creamy), EWS, and General quotas.	P0 Core	`feat_statutory_rosters`
F134	**CBSE LOC Registration Formatter**	Generates Class 10/12 Board exam registration sheets matching CBSE standard LOC format.	P0 Core	`feat_cbse_loc`
F549	**DPDPA Minor Consent & Erasure**	Verifiable parental consent logging and statutory "Right to be Forgotten" erasure workflow.	P0 Core	`feat_dpdpa_compliance`
F272	**Statutory Coaching Refund Calculator**	Enforces MoE 2024 mandatory 10-day pro-rata refund calculation for student course withdrawal.	P0 Core	`feat_fee_refund`

MODULE 8 • CLASSROOM OPERATIONS • CORE INNOVATION

Classroom Roll Call: The True Offline-First Sync Engine

Empowering educators to complete roll call in under 30 seconds with 0ms UI latency, complete offline durability, and automatic background sync.

⚠ The Internet Problem in Generic ERPs

- ✕ **Wi-Fi Dependent Failure**: In Indian schools, classrooms in basements or rural areas have spotty connectivity. Cloud ERPs display loading spinners or connection timed out.
- ✕ **Wasted Teaching Time**: Calling roll takes 15 minutes of lecture time due to slow network round-trips for every student attendance checkbox.
- ✕ **Data Loss on Submit**: If network drops while clicking "Submit", the entire attendance sheet is wiped out, forcing teachers back to paper registers.
- ✕ **Delayed Parent Communication**: Absence notices sent 6 hours late, defeating student safety purposes.

✦ Our Dexie.js Offline Outbox Sync Engine

- ✓ **Zero-Latency Local Storage**: Students render instantly from browser IndexedDB. "Mark All Present" → tap 2 absent students → done in 15 seconds.
- ✓ **Durable Outbox Queue**: Offline records save locally with UUID idempotency keys; UI indicates: *"Saved locally (Offline - 1 pending sync)"*.
- ✓ **Seamless Auto-Sync**: When device reconnects to Wi-Fi/4G, the background sync engine pushes queued mutations to `/api/sync/outbox` with 0 data loss.
- ✓ **Instant 15-Minute WhatsApp Alert**: Server sync immediately enqueues automated WhatsApp/DLT SMS notifications to parents of absent students.

Attendance & Roll Call Feature Catalogue (Sample)

Module 8 Full Specifications

ID	FEATURE NAME	TECHNICAL FUNCTION & CAPABILITY	PRIORITY	FEATURE FLAG KEY
F181	**Daily Classroom Roll Call**	High-speed UI: 1-click "Mark All Present", tap to toggle Absent/Late with 0ms UI response.	P0 Core	`feat_daily_attendance`
F182	**True Offline Roll Call Sync**	Marks roll call completely offline in IndexedDB; auto-syncs via durable Outbox queue.	P0 Core	`feat_offline_rollcall`
F183	**Period / Subject-Wise Attendance**	Higher-ed period-by-period roll call for individual college lecture sessions.	P0 Core	`feat_period_attendance`
F184	**Real-Time Absent SMS to Parents**	Auto-dispatches WhatsApp/SMS to guardian within 15 mins of roll call finalization.	P0 Core	`feat_absent_parent_sms`
F187	**75% Mandatory Attendance Rule**	Visual warning badges for students falling below the statutory exam debarment line.	P0 Core	`feat_attendance_threshold`
F188	**Debarred Student Admit Card Lock**	Automatically blocks examination hall ticket printing if attendance is below 75%.	P0 Core	`feat_debarment_block`
F189	**Monthly Attendance Muster Register**	Classic Indian physical register view with P/A monthly grid and aggregate totals.	P0 Core	`feat_monthly_register`

MODULE 9 • ACADEMIC SCHEDULING • OPERATIONS

Master Timetable, Scheduling Matrix & Instant Substitutions

Clash-free weekly drag-and-drop classroom scheduling with room capacity enforcement and automated 1-click substitute teacher allocation.

⚠ Scheduling Chaos in Generic ERPs

- ✕ **No Real-time Clash Detection**: Allows scheduling the same physics teacher to two different classes in the same period without warning.
- ✕ **The "Morning Substitution" Nightmare**: When 3 teachers call in sick at 7:30 AM, the Vice Principal spends 45 minutes manually scanning paper timetables to find proxies.
- ✕ **Lab Block Limitations**: Cannot model continuous 2-hour or 3-hour practical science lab blocks across regular periods.
- ✕ **Unbalanced Workloads**: Senior faculty assigned 32 periods a week while junior faculty have only 14, causing staff dissatisfaction.

✦ Our Intelligent Scheduling & Proxy Engine

- ✓ **Visual Drag-and-Drop Matrix**: Instant clash detection: flags teacher or room collision before the slot is even saved.
- ✓ **1-Click Substitution Engine**: When a teacher logs leave, system instantly scans schedules and suggests available faculty ranked by subject match.
- ✓ **Instant Proxy WhatsApp Alert**: Assigned substitute receives an immediate notification: *"You have proxy duty for Class 10-A, Period 3 in Room 204"*.
- ✓ **Weekly Workload Balancer**: Visual dashboard ensuring every full-time teacher has a balanced 24-28 periods per week per CBSE/UGC norms.

Timetable & Substitution Feature Catalogue (Sample)

Module 9 Full Specifications

ID	FEATURE NAME	TECHNICAL FUNCTION & CAPABILITY	PRIORITY	FEATURE FLAG KEY
F216	**Master Class Timetable Grid**	Weekly visual grid (Mon-Sat, Periods 1-8) with drag-and-drop slot allocation.	P0 Core	`feat_timetable_grid`
F217	**Teacher Individual Timetable View**	Dedicated schedule view per faculty member showing free periods and duty halls.	P0 Core	`feat_teacher_timetable`
F220	**Real-Time Conflict Detection**	Validation engine preventing double-booking of teacher, room, or lab equipment.	P0 Core	`feat_timetable_conflict_check`
F221	**1-Click Substitution Engine**	Scans schedules and recommends free eligible subject teachers during colleague sick leave.	P0 Core	`feat_substitution_engine`
F222	**Instant Substitution WhatsApp Alert**	Dispatches immediate mobile alert to substitute teacher with period and room details.	P0 Core	`feat_proxy_alert`
F223	**Printable Master Timetable PDF**	High-resolution A3/A4 master schedule generator for the staffroom noticeboard.	P0 Core	`feat_timetable_print`
F226	**Weekly Faculty Workload Balancer**	Bar chart ensuring teachers maintain balanced 24-28 periods/week per UGC/CBSE rules.	P0 Core	`feat_workload_balance`

MODULE 10 & 11 • FINANCIAL LEDGER • INVOICING

Fee Structures, Dynamic NPCI UPI QR & POS Receipts

Double-entry student fee ledger, automated concession rules, dynamic 0% commission UPI QR invoices, and fast cashier counter POS printing.

⚠ Financial Bottlenecks in Generic ERPs

- ✕ **High Gateway Commissions**: Schools lose 1.5% to 2% on every fee transaction through standard gateway credit card / netbanking pipes.
- ✕ **Long Cashier Queues**: Cashier takes 3 to 5 minutes per parent on busy fee submission days due to complex form loading.
- ✕ **Manual Concession Math**: Clerks calculate sibling discounts and late fines manually on physical calculators, prone to human error.
- ✕ **Cheque Reconciliation Nightmare**: Unclear tracking of deposited vs bounced cheques, causing ledger discrepancies.

✦ Our Zero-Commission Financial Engine

- ✓ **Dynamic NPCI UPI QR Code**: Every invoice renders a BharatPe/NPCI UPI QR. Parents scan via GPay/PhonePe; settles directly with **0% MDR**.
- ✓ **30-Second POS Cash Counter**: Search student → select due terms → collect cash → 1-click print branded 80mm thermal receipt.
- ✓ **Automated Sibling & Staff Concessions**: Rules engine automatically applies 25% on 2nd child and 50% on 3rd child tuition fees.
- ✓ **GST Compliant Invoices (SAC 9992)**: Automatic 0% GST exemption for schools vs 18% GST calculation for commercial coaching academies.

Fees, Invoicing & Payment Feature Catalogue (Sample)

Module 10 & 11 Full Specifications

ID	FEATURE NAME	TECHNICAL FUNCTION & CAPABILITY	PRIORITY	FEATURE FLAG KEY
F246	**Flexible Fee Head Master**	Defines heads: Tuition, Admission, Term, Lab, Library, Transport, and Uniform.	P0 Core	`feat_fee_heads`
F250	**Daily / Fixed Late Fine Rules**	Computes ₹50/day late fine or 5% flat charge automatically after grace period cutoff.	P0 Core	`feat_late_fines`
F251	**Sibling Concession Engine**	Auto-applies 20% discount on 2nd sibling and 50% on 3rd sibling tuition fee.	P0 Core	`feat_sibling_concession`
F267	**1-Click WhatsApp Defaulter Broadcast**	Batch dispatches personalized payment links to parents with unpaid balances over 30 days.	P0 Core	`feat_whatsapp_fee_reminder`
F281	**Fast POS Cash Counter Collection**	Sub-30 second checkout: search student, select terms, collect, and print receipt.	P0 Core	`feat_pos_fee_counter`
F282	**Dynamic NPCI UPI QR on Invoices**	Renders dynamic BharatPe/NPCI UPI QR code on student fee bill for zero-commission payment.	P0 Core	`feat_dynamic_upi_qr`
F287	**Branded 80mm Thermal Receipt Print**	Standard 3-inch thermal printer output with school logo, receipt no, and verification QR.	P0 Core	`feat_thermal_receipt_print`

MODULE 12 • EVALUATION • ACADEMIC RECORDS

Examinations, CBSE Grading, Broadsheet & Report Cards

Fast batch mark entry with keyboard navigation, CBSE 9-point & UGC SGPA/CGPA engines, A3 tabulation broadsheets, and digital report cards.

⚠ Grading Frustrations in Generic ERPs

- ✕ **Painful Slow Mark Entry**: Teachers must click, type, and click save for every individual student; entering 60 students takes 45 minutes.
- ✕ **Incompatible Grading Rules**: Cannot calculate the CBSE 9-point scale (A1 to E) or University SGPA/CGPA formulas correctly.
- ✕ **No Broadsheet Overview**: School leaders cannot see an A3 master tabulation sheet showing all students, all subjects, totals, and ranks together.
- ✕ **Plain Unattractive Report Cards**: Produces ugly black-and-white printouts without student photos, co-scholastic rubrics, or verification QR codes.

✦ Our High-Speed Evaluation Platform

- ✓ **Excel-Like Keyboard Mark Entry**: Enter scores using down-arrow or Enter key; auto-validates against max marks with 0ms calculation latency.
- ✓ **Native CBSE & University Engines**: Pre-built algorithms for CBSE 9-point percentiles and UGC 10-point semester credit SGPA/CGPA.
- ✓ **A3 Class Tabulation Broadsheet**: Master overview of all students and subjects; exportable to Excel (.xlsx) for examination board records.
- ✓ **Stunning Digital Report Cards**: Branded PDF generation with student photo, scholastic marks, co-scholastic grades, remarks, and QR code.

Examination & Evaluation Feature Catalogue (Sample)

Module 12 Full Specifications

ID	FEATURE NAME	TECHNICAL FUNCTION & CAPABILITY	PRIORITY	FEATURE FLAG KEY
F316	**Exam Term & Assessment Master**	Defines Formative (FA), Summative (SA), Mid-Term, Pre-Board, and Final examination terms.	P0 Core	`feat_exam_master`
F318	**Fast Batch Mark Entry Table**	Excel-like grid: Enter key advances down, auto-tabulating totals, percentages, and grades.	P0 Core	`feat_batch_marks_entry`
F320	**CBSE 9-Point Grading Scale (A1-E)**	Automatic conversion of subject scores to official CBSE letter grades based on percentiles.	P0 Core	`feat_cbse_grading`
F321	**University SGPA & CGPA Engine**	Semester Grade Point Average and Cumulative GPA per official UGC 10-point credit scale.	P0 Core	`feat_sgpa_cgpa_engine`
F323	**Class Tabulation Sheet (Broadsheet)**	Master A3 overview showing all students, all subjects, totals, ranks, and pass/fail status.	P0 Core	`feat_tabulation_sheet`
F324	**Branded Report Card PDF Generator**	Official report card generator with school crest, student photo, attendance, and QR code.	P0 Core	`feat_report_card_generator`
F329	**Exam Hall Ticket / Admit Card Creator**	Generates print-ready student admit cards with exam dates, room numbers, and photo barcode.	P0 Core	`feat_hall_ticket_creator`

MODULES 13–23 • OPERATIONS • ENGAGEMENT • PORTALS

Omnichannel Communication, Operations & Portals

WhatsApp Business API broadcasts, Library barcode circulation, Transport fleet GPS, dedicated mobile portals, and Universal Command Palette.

⚠ Operational Gaps in Generic ERPs

- ✕ **Disconnected Third-Party Add-Ons**: School must buy separate software for library barcodes, bus GPS tracking, and WhatsApp messaging.
- ✕ **Teacher Privacy Violation**: Teachers forced to share personal phone numbers on informal WhatsApp groups, receiving late-night calls.
- ✕ **Weak Parent Portal**: Parents cannot easily switch between 2 children studying in different grades without logging out.
- ✕ **Zero Keyboard Acceleration**: Staff waste hours clicking through complex submenus to find simple student records.

✦ Our Integrated Operational Ecosystem

- ✓ **Unified Communications Hub**: Official WhatsApp Business API + TRAI DLT SMS with pre-approved template verification.
- ✓ **Teacher Privacy Masking & DND**: Parents message teachers through the in-app portal; teacher personal mobile is 100% masked with 6 PM Quiet Hours.
- ✓ **Multi-Child Parent Portal**: Switch between siblings across Class 10 and Class 6 in 1 tap with unified 1-tap fee payment.
- ✓ **Universal Command Palette (Ctrl + K)**: Press `Ctrl + K` anywhere to jump to any student, invoice, or setting in 2 keystrokes.

Operations, Communication & Portal Feature Catalogue (Sample)

Modules 13–23 Full Specifications

ID	FEATURE NAME	TECHNICAL FUNCTION & CAPABILITY	PRIORITY	FEATURE FLAG KEY
F383	**WhatsApp Business API Broadcast**	Dispatches official verified template notifications for fee receipts and urgent notices.	P0 Core	`feat_whatsapp_broadcast`
F388	**Teacher Phone Privacy Protection**	In-app parent-teacher messaging completely masks teacher personal mobile phone numbers.	P0 Core	`feat_teacher_privacy`
F413	**10-Second Library Barcode Checkout**	Scan student ID barcode → scan book ISBN barcode → checkout completed in seconds.	P0 Core	`feat_book_checkout`
F433	**Student Bus Route Allocation**	Maps student to specific bus stop and auto-calculates distance fee slabs in fee ledger.	P0 Core	`feat_student_bus_allocation`
F452	**Hostel Room & Bed Allocation**	Visual room/bed allocation matrix (Single, AC, Dormitory) with curfew roll call.	P0 Core	`feat_hostel_room_allocation`
F492	**Multi-Child Parent Switcher**	Parents with multiple enrolled children switch between student profiles with a single tap.	P0 Core	`feat_multichild_parent`
F566	**Universal Command Palette (Ctrl + K)**	Instant spotlight search to jump to any student, class, invoice, or setting in 2 keystrokes.	P0 Core	`feat_command_palette`